VELLORE INSTITUTE OF TECHNOLOGY (VIT), CHENNAI · SCHOOL OF ADVANCED SCIENCES

Can a Chi-Square Test Catch Fake Numbers?

Detecting Data Fabrication Using Benford's Law and Pearson's Goodness-of-Fit Analysis

Course: BMAT202 – Probability and Statistics (DA3)
Syllabus Module: Module 6 – Hypothesis Testing-II (Chi-Square & Goodness-of-Fit)
Investigative Team 14: Pranav S. (24BCE1166) & Team Members
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01 REAL-LIFE PROBLEM & SIGNIFICANCE

In modern forensic accounting, tax audit compliance, and research verification, manual review of millions of financial or experimental transactions is impossible. Fraudsters routinely fabricate numbers to cover shortfalls or pad expenses.

Core Psychological Conflict: Natural growth processes follow logarithmic scales, whereas humans inventing numbers instinctively distribute digits uniformly (~11.1% each) and avoid repeating digit 1.

Objective: Leverage an undergraduate statistical hypothesis test (χ^2 goodness-of-fit) to establish an automated, non-invasive triage tool for detecting fabricated numerical records.

02 CONCEPTS & HYPOTHESIS TESTING

Benford's First-Digit Law: In naturally occurring unconstrained data spanning multiple orders of magnitude, the probability of leading digit $d \in \{1, ..., 9\}$ is:

$$P(d) = \log_{10}(1 + 1/d)$$

Digit 1: 30.1% | Digit 2: 17.6% | Digit 3: 12.5% ... Digit 9: 4.6%

Statistical Hypotheses (Module 6):

- H_0: First-digit distribution follows Benford's Law.
- H_1: First-digit distribution departs from Benford's Law.

Pearson's Chi-Square Test Statistic:

$$\chi^2 = \sum_{i=1}^{9} (O_i - E_i)^2 / E_i, \quad E_i = n \cdot P(i)$$

Degrees of Freedom: df = k - 1 - m = 9 - 1 - 0 = **8** (no parameters estimated from sample).
Critical Value ($\alpha = 0.05$, df = 8): $\chi^2_{crit} = 15.507$.

Cochran's Condition: $E_i \geq 5$ for all bins &implies; n ≥ 110. World Bank n = 217 yields $E_9 = 9.93 > 5.0$ (Passed).

03 AUDIT SCREENING WORKFLOW

1. **Suitability & Boundary Check**
   Verify positive, ratio scale, ≥3 orders of magnitude (no exam scores / IDs).
2. **Digit Extraction Algorithm**
   Standardized formatC(x, "e") isolates leading digit d ∈ {1..9} without error.
3. **Dual Goodness-of-Fit Evaluation**
   Calculate χ^2 (df=8, p-val) + scale-free Mean Absolute Deviation (MAD).
4. **Automated Risk Triage**
   Green ($\chi^2 \leq 15.5$) • Yellow (boundary review) • Red ($\chi^2 > 15.5$, high risk).
5. **Residual-Driven Audit Sampling**
   Focus substantive manual audit on digit bins with residuals $R_i > +2.0$.

04 EMPIRICAL DATA PROVENANCE

Dataset A: Genuine Demographic Data

- **Source:** World Bank 2023 World Development Indicators (`SP.POP.TOTL`).
- **Cleaning:** 266 entities → excluded 1 null & 48 regional aggregates → **n = 217 sovereign nations**.
- **Span:** 9,816 (Tuvalu) to 1.44B (India) — spans 5 orders of magnitude without artificial caps.

Dataset B: Simulated Fabricated Example

- **Sample:** n = 217, fixed seed = 2026.
- **Cognitive Mechanism:** Models the *equal-likelihood fallacy* where humans invent numbers choosing digits 1–9 roughly uniformly (~11.1% each) across commercial invoice scales.
- *Disclaimer: Demonstration example; not actual fraud allegations.*

05 EMPIRICAL FINDINGS & VISUALIZATIONS

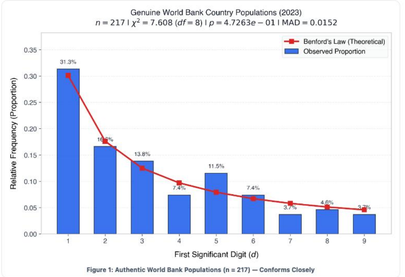


Figure 1: Authentic World Bank Populations (n = 217) — Conforms Closely

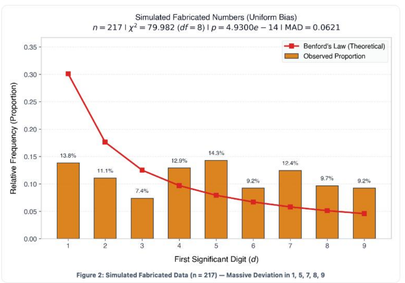


Figure 2: Simulated Fabricated Data (n = 217) — Massive Deviation in 1, 5, 7, 8, 9

Dataset	n	χ^2	p-value	MAD	Decision (α=0.05)
World Bank (Genuine)	217	7.608	0.4726	0.0152	**Fail to Reject H_0**
Simulated Fabrication	217	79.982	4.93e-14	0.0621	**Reject H_0 (Flagged)**
R Rivers Benchmark	141	44.177	5.27e-07	0.0525	**Reject H_0 (Truncated)**

06 THE RIVER LENGTHS PARADOX

The historical North American rivers dataset (n = 141) failed Benford's test (χ^2 = 44.18, p < 0.001). Why?

Selection Truncation Effect: The USGS World Almanac recorded only "major" rivers (>135 miles). Omitting small streams broke scale invariance, clustering values between 200–400 miles (digit 3 count: 36 vs exp 17.6).

Vital Lesson: A rejection is NOT proof of fraud! Bounded or truncated authentic datasets naturally depart from Benford.

07 CONCLUSIONS & AUDIT LIMITS

- **Clear Contrast:** Genuine data conformed (χ^2 = 7.608, p = 0.473), whereas uniform fabrication failed overwhelmingly (χ^2 = 79.982, p = 4.93×10^{-14}).
- **Effect Size is Mandatory:** With large samples, χ^2 rejects tiny deviations. Mean Absolute Deviation (MAD) provides scale-free practical significance.
- **Audit Ethics:** Rejection flags anomalies for manual review; failure to reject does not guarantee innocence.

08 VERIFIED REFERENCES

- **Benford, F. (1938).** Law of Anomalous Numbers. *Proc. Amer. Phil. Soc.*, 78(4).
- **Nigrini, M. J. (2012).** *Benford's Law: Forensic Accounting & Fraud Detection*. Wiley.
- **Cochran, W. G. (1954).** Strengthening Common χ^2 Tests. *Biometrics*, 10(4).
- **Hill, T. P. (1995).** Statistical Derivation of Significant-Digit Law. *Stat. Sci.*, 10(4).
- **World Bank (2024).** World Development Indicators: Total Population (SP.POP.TOTL).